

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from 3,900 transactions to guide strategic business decisions.

Dataset at a Glance

3,900

Purchases

Transactional records across all product categories

18

Features

Demographics, purchase details, and shopping behavior

37

Missing Values

Only in Review Rating column, imputed by category median

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount, Promo Code, Frequency, Review Rating, Shipping Type).

Data Preparation in Python

Cleaning & Standardization

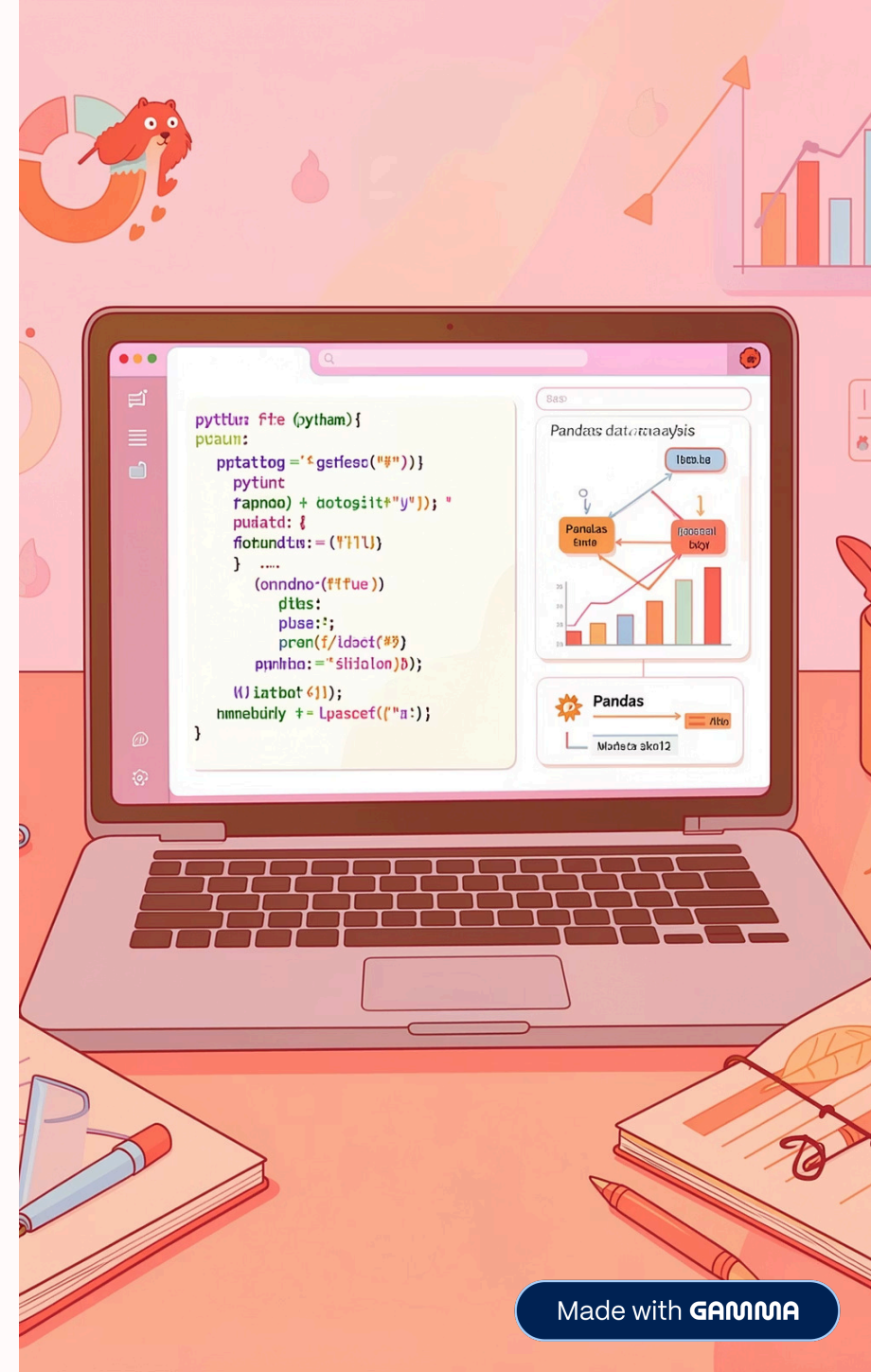
Loaded data via pandas, checked structure with `df.info()`, imputed missing Review Ratings using category medians, and renamed columns to snake_case.

Feature Engineering

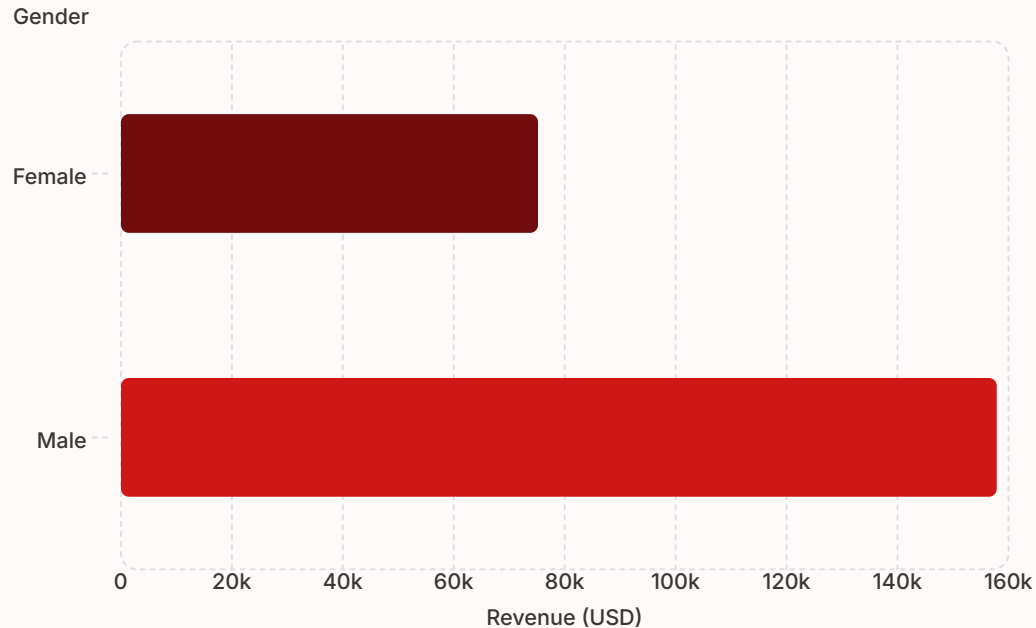
Created `age_group` by binning customer ages and `purchase_frequency_days` from purchase data. Dropped redundant `promo_code_used` column.

Database Integration

Connected Python to PostgreSQL and loaded the cleaned DataFrame for SQL-based analysis.



Revenue by Gender



Key Finding

Male customers generated **\$157,890** in revenue — more than double the **\$75,191** from female customers. This significant gap highlights a key opportunity to better engage and retain female shoppers through targeted campaigns and product offerings.

Products, Ratings & Discounts

Top 5 Products by Rating

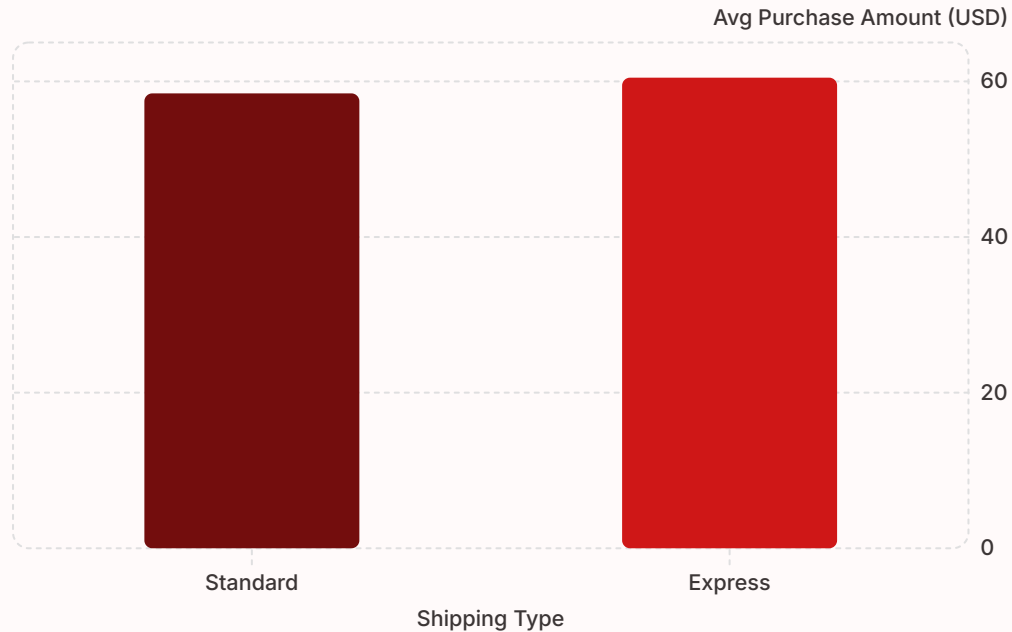
Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Rates

Item	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Hats, Sneakers, and Coats see the highest discount rates (~47–50%), suggesting strong price sensitivity in these categories. Meanwhile, Gloves and Sandals lead in customer satisfaction with ratings above 3.84.

Shipping & Subscription Insights

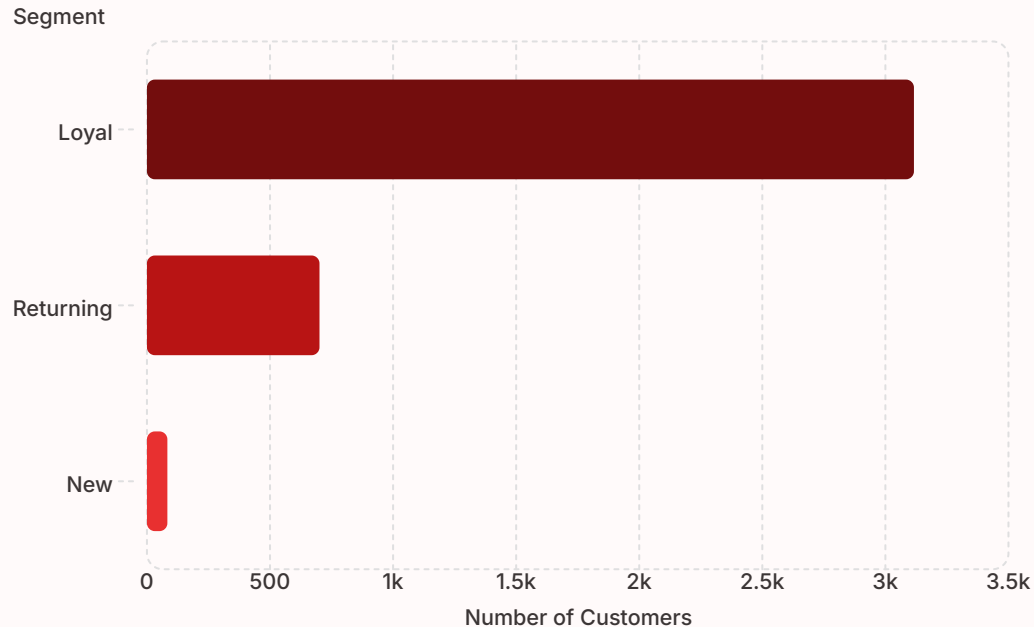


Subscribers vs. Non-Subscribers

Status	Customers	Revenue
Yes	1,053	\$62,645
No	2,847	\$170,436

Express shipping users spend slightly more on average (**\$60.48** vs. **\$58.46**). Non-subscribers dominate in both customer count and total revenue, representing a major opportunity for subscription conversion.

Customer Segmentation



Segment Breakdown

Customers were classified into three segments based on purchase history:

- **Loyal (3,116):** The dominant segment, representing the core customer base
- **Returning (701):** Customers with repeat purchases but not yet loyal
- **New (83):** First-time buyers — a small but important growth opportunity

Of customers with more than 5 purchases, 958 are subscribers and 2,518 are not, indicating room to convert repeat buyers into subscribers.

Top Products by Category

Accessories

1. Jewelry — 171 orders
2. Sunglasses — 161 orders
3. Belt — 161 orders

Clothing

1. Blouse — 171 orders
2. Pants — 171 orders
3. Shirt — 169 orders

Footwear

1. Sandals — 160 orders
2. Shoes — 150 orders
3. Sneakers — 145 orders

Outerwear

1. Jacket — 163 orders
2. Coat — 161 orders



Power BI Dashboard



Key Metrics & Visuals

KPIs

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Filters

Subscription, Gender, Category,
Shipping Type

Visualizations

Revenue & Sales by Category and
Age Group; Subscription donut
(27% Yes, 73% No)

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.



Loyalty Programs

Reward repeat buyers to move them into the high-value "Loyal" segment.



Review Discount Policy

Balance sales boosts with margin control, especially on high-discount items like Hats and Sneakers.



Targeted Marketing

Focus campaigns on high-revenue age groups, express-shipping users, and top-rated products.

